

LLMs, bias and fairness

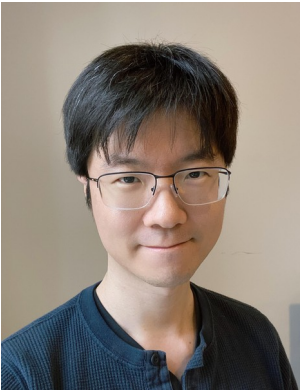
Definizioni

- Bias (pregiudizio): *“inclination or prejudice for or against one person or group, especially in a way considered to be unfair”* (Oxford Languages)
- Fairness (equita’/imparzialità): *“impartial and just treatment or behaviour without favouritism or discrimination”* (Oxford Languages)

Google Scholar “Ilm bias” search

- Returns 257,000 ‘hits’
- Page 10 of results, 9/10 articles on target (one on cognitive bias in using LLM)
- E c’e’ pure il gatto che si morde la coda ...
“Investigating Bias in LLM-Based Bias Detection
: Disparities between LLMs and Human Perception” (COLING 25)

Unveiling Performance Challenges of LLMs in Low-Resource Healthcare: A Demographic Fairness Perspective (COLING 25)



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Introduction

Research Questions: Will LLMs be as successful in healthcare as they have been in other domains? Will they help mitigate health disparities and achieve health equity?

Challenges:

- Breadth of domain knowledge
- Limited data resources with demographic information
- Inherent ethical considerations



Introduction: Motivations

✧ Lack of comprehensive evaluation of LLMs on real-world healthcare tasks:

- we propose 6 benchmarks and evaluate performance

✧ Concerns regarding biases towards certain groups and potential to reinforce health disparities:

- analysis from **demographic fairness** perspective



Four Datasets, Six Benchmarks

1. Medical Question Answering (US Medical Licensing Examination) [Jin et al 2021]

2. Patient goal achievement prediction from Health Coaching Datasets [Gupta et al 2020, Zhou et al 2024]

3+4. One-year mortality prediction and 90-day readmission - from MIMIC-IV [Johnson et al 2020]

5+6. Schizophrenia/Bipolar Disorder/Healthy prediction from spoken dialogues:

Meeting New Neighbor and *Complaining to a Landlord* [Aich et al 2022]

MIMIC-Mortality

Chief Complaint:

Abdominal pain and decreased ostomy...

History of Present Illness:

...with a complicated surgical history..had a discussion on surgical options due to her recurrent SBOs...

Past Medical History:

Ulcerative colitis s/p total abdominal colectomy at...

Demographics:

{Age: 68, Race: White, Gender: Female}

Label (1-year mortality): *Alive*

Bipolar and SCZ Interviews (Neighbor)

Dialogue:

Participant: Yes, I was. I was just coming to uh.. see how you're doin' today. Do you have a name?

Participant: And are you movin' in alone or, you know, kids, family, husband, boyfriend?

Interviewer: Uh.. nope. Just- just me and my dog.

Participant: All right, just you and your dog. Oh, okay. You're pet lover..

Label: *Schizophrenia*



Dataset Statistics

Task	Fine-Tuning	Test
MIMIC-Mortality	5000	500
MIMIC-Readmission	5000	500
MedQA	5000	175
Health Coaching	120	60
SCZ and Bipolar: Neighbor	190	261
SCZ and Bipolar: Landlord	188	261



Three approaches, three LLMs

- In-Context Learning (ICL) with Chain-of-Thought
 - OpenAI GPT-4
 - Claude-3 (Sonnet)
 - LLaMA-3 (8b)
- Parameter Efficient Fine-Tuning (PEFT) (LoRA)
 - LLaMA-3 (8b)
- LLM as Agents (ReAct and Reflexion) [external knowledge]
 - OpenAI GPT-4

Prompt example:

Mortality / Readmission: Assess the 1-year mortality risk (low or high) / 90-day readmission chance for the patient based on the given de-identified clinical note. Using medical evidence only. If uncertain, provide your best evidence-based estimate.

[Note]

Chief complaint / history of present illness / past medical history / lab results.



LLM as Agents

**Simplified
“LLM as Agents”
Framework**



Results: Accuracy

Baselines	Backbones	Mortality (Mimic)	MedQA	SCZ/Bipolar (Neighbor)	SCZ/Bipolar (Landlord)
Random Guess	-	50.0	20.0	33.3	33.3
Chain-of-Thought	GPT-4	77.0 (79.0)	(68.6)	41.0 (39.1)	28.2 (36.8)
Chain-of-Thought	Claude-3	72.6 (50.8)	(65.7)	37.5 (38.3)	37.2 (34.1)
Chain-of-Thought	LLaMA-3	73.0 (72.2)	(59.4)	33.0 (33.3)	34.1 (33.7)
Supervised Fine-Tuning	LLaMA-3	68.2	71.4	49.4	41.4
LLM as Agent	GPT-4	66.7	86.9	36.8	36.8

- LLMs perform so so on real-world healthcare tasks: some models barely surpass random guess
- While often close-source > open-source, best framework varies by tasks (e.g. Mortality vs SCZ/Bipolar)
- Agent approach works for MedQA, fails real-world cases
- Demographics prompting shows mixed results



Two Metrics for Fairness

[Zemel et al 2013, Liu et al 2023]

Demographic Parity Difference (DPD) tests whether model treats various subgroups similarly

$$DPD = P(\hat{Y} = 1 \mid Z = z_i) - P(\hat{Y} = 1 \mid Z \neq z_i)$$

$\hat{Y}$ = outcome, Z = protected attribute (demographic information)

Does not consider ground truth



Two Metrics for Fairness

Equal Opportunity Difference (EOD)

measures whether different subgroups have equal likelihood of being accurately classified

$$EOD = P(\hat{Y} = 1 \mid Y = 1, Z = z_i) - P(\hat{Y} = 1 \mid Y = 1, Z \neq z_i)$$

$\hat{Y}$ = outcome, Y = ground truth, Z = protected attribute (demographic information)



Fairness – Predict Mortality

Setting	DPD		EOD	
	AfrAm - White	Fem - Male	AfrAm - White	Fem - Male
COT-GPT-4	-8.2 (-7.4)	-5.1 (-2.7)	-2.5 (-4.4)	-4.6 (-5.3)
COT-Claude-3	-11.5 (-6.4)	-8.7 (+1.5)	-3.8 (-3.8)	-4.5 (-7.2)
COT-LLaMA-3	-8.3 (-17.6)	-2.8 (-3.0)	-4.2 (-8.0)	-5.9 (-5.2)
FinTun LLaMA-3	-14.3	-12.7	-9.4	-5.3
Agent (GPT-4)	+2.1	-2.1	+2.4	+2.8

Unfairness present across all tasks, more in race than gender

- African American patients consistently receive less favorable predictions
- Fine-tuning improves fairness for some tasks (readmission) but worsens others (mortality, health coaching)



Additional Experiments: **Demographic Awareness and Bias in Conversation**

Accuracy in predicting patient's race in conversational data:

	Health Coaching	SCZ/Bipolar (Neighbor)	SCZ/Bipolar (Landlord)
Random	33.3	50.0	50.0
GPT-4	Refusal	75.3	78.8
Claude-3	Refusal	Refusal	Refusal
LLaMA-3	40.0	59.6	61.7

Ask for explanation: Trite stereotypes in LLaMA reasoning (in paper appendix)

Conclusions & Future Work

- LLMs are not as performant in real-world healthcare tasks
- Substantial differences in Demographic Parity and more importantly in Equal Opportunity

Need: more healthcare data with demographic information

Big question: how to mitigate bias and improve fairness?



References (COLING 25)

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Additional slides



6 Datasets and Task Formulations



Results: Accuracy

Baselines	Backbones	Mortality	Readmission	MedQA	HealthCo	Neighbor	Landlord
Random Guess	-	50.0	50.0	20.0	50.0	33.3	33.3
ICL/Chain-of-Thought	GPT-4	77.0 (79.0)	55.3 (56.8)	-(68.6)	76.7 (76.7)	41.0 (39.1)	28.2 (36.8)
ICL/Chain-of-Thought	Claude-3	72.6 (50.8)	52.9 (57.6)	-(65.7)	80.0 (80.0)	37.5 (38.3)	37.2 (34.1)
ICL/Chain-of-Thought	LLaMA-3	73.0 (72.2)	53.3 (55.3)	-(59.4)	70.0(70.0)	33.0 (33.3)	34.1 (33.7)
Supervised Fine-Tuning	LLaMA-3	68.2	48.2	71.4	70.0	49.4	41.4
LLM as Agent	GPT-4	66.7	53.3	86.9	70.0	36.8	36.8

- LLMs struggle with real-world healthcare tasks (Some barely surpass random guess)
- While generally close-source > open-source, best framework varies by tasks
- Agent approach works for MedQA, fails real-world cases
- Demographics prompting shows mixed results



Results: MedQA Accuracy

Baselines	Accuracy			
	African-American	Asian	Hispanic	White
COT-GPT-4	62	60	68	84
COT-Claude-3	64	62	60	74
COT-LLaMA-3	60	56	52	66
SFT (LLaMA-3)	86	62	52	76
Agent (GPT-4)	86	86	88	88

- Discrepancy in in-context learning performance in MedQA targeting different racial groups
- LLM-as-Agent trivially works well since it can access answer from internet